MILITARY SPECIFICATION

SOURCES, LED, FIBER OPTIC, 820-910 NANOMETERS WAVELENGTH RANGE, GLASS PIGTAILED TYPE (METRIC)

This specification is approved for use by Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for pigtailed LED (light emitting diode) devices operating in the wavelength range from 820 to 910 nanometers and compatible with fiber optic cable as specified in DOD-C-85045. Both hermetic and nonhermetic pigtail configured LEDS (product assurance quality levels A and B, respectively) may be qualified to this specification if they submit to and pass 100 percent screening tests as specified herein. Pigtail connectorization for LEDS to be qualified shall be as specified.

1.2 Classification. Source classification shall be in accordance with DOD-S-24622. Standard sources to be acquired by this specification shall be as specified in table I.

TABLE I. Source characteristics and part numbers.

Military part number D 24622/1-	Sealing level	Wavelength of operation	Pigtail fiber	
			Core/cladding diameter (µm)	Core index profile
A 08 B B 08 B	fully hermetic nonhermetic	820 - 910 nanometers	50/125	graded
A 08 C B 08 C	"	"	100/140	step

NOTE: The source covers the wavelength band from 820 and 910 nanometers, but its performance criteria shall be met at the specified center wavelength. The center wavelength specified is the center operating wavelength of the LED source as specified in DOD-S-24622.

1.2.1 Part number. The part number shall be in accordance with DOD-S-24622 (see table I).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Sea Systems Command (SEA 55Z3), DOD Standardization Program and Documents Division, Department of the Navy, Washington, D.C. 20362, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:24Z
Check the source to verify that this is the current version before use.

1.3. Design maximum and minimum ratings.

Minimum continuous forward current (I_{FM}) at T_A = 25°C - - - - - 150 mA

(Derate I_{FM} above T_A = 50°C at 2 mA/°C).

Minimum peak forward voltage (I_{FRM}) at T_A = 25°C, DF = $\frac{tp = 1\ \mu s}{0.1}$ 1.0 A

Minimum peak reverse current (V_{RM}) at T_A = 25°C, I_{RM} = 10 µA- - - 3 V

Storage and operating temperature range:

Hermetic pigtailed LEDS (Quality level A)

Operating- - - - - - - - - - - - - T_C = -20°C to +100°C

Storage- - - - - - - - - - - - - - T_C = -40°C to +125°C

Nonhermetic pigtailed LEDS (Quality level B)

Operating- - - - - - - - - - - - - T_C = -20°C to +70°C

Storage- - - - - - - - - - - - - - T_C = -40°C to +85°C

External lead maximum solder temperature (5 seconds)- - - 250°C

Pigtail assembly minimum bend radius - - - - - - - - - - - 10 mm

Pigtail assembly minimum pull strength - - - - - - - - - 20 N

Pigtail fiber minimum pull strength - - - - - - - - - - - 5 N

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Government specification and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

DOD-S-24622 - Sources, LED, Fiber Optic, General Specification for.

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

MIL-STD-750 - Test Methods for Semiconductor Parts.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:24Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with DOD-S-24622 and as specified herein.

3.2 Abbreviations and symbols. The abbreviations and symbols used herein shall be as specified in MIL-S-19500.

3.3 Design, construction and physical dimensions. LEDS shall be of the design, construction and physical dimensions shown on figure 1.

3.3.1 Lead conductors.

3.3.1.1 Material and finish. Anode and cathode lead conductors shall be Alloy-52 or Kovar of diameter 0.51 millimeter (mm) (American Wire Guage 24) with a hot solder dip finish. Conductors shall be diametrically opposed and displaced an equal distance from the LED assembly center-line axis (see symbol M, figure 1, for the conductor spacing).

3.3.1.2 Polarity identification. A white stripe, 1 mm wide and 13.2 mm long (see dimension symbol C, figure 1) shall be affixed to the assembly case, collateral and parallel with the cathode conductor, to identify it.

3.3.2 LED case assembly. The LED case and stud assembly shall be gold plated with a visually smooth surface finish.

3.3.2.1 Mounting. The case assembly shall be mounted directly to a suitable copper heat sink by means of the threaded stud/nut arrangement used. Heat conductive material (lubricant or metal-loaded paste) shall be used between the case and heat sink for efficient thermal transfer. The case shall be operated at electrical ground and no heat sink electrical isolation is required.

3.3.2.2 Color. The source case (excluding the stud) shall be painted black with a flat finish.

3.3.3 Pigtail assembly.

3.3.3.1 Sheath. The fiber pigtail sheath material and construction shall provide the bend-restrictive property which prevents the pigtail assembly from exceeding the minimum bend radius specified in 1.3. Sheath materials shall be nonmetallic.

3.3.3.2 Strain relief. Strain relief shall be provided between the case body and the pigtail sheath. Metallic or nonmetallic materials may be used for strain relief construction. The overall length of the strain relief member shall not exceed 30 mm and shall provide a minimum bend radius of 20 mm under normal usage.

3.3.3.3 Fiber. The pigtail fiber selected shall contain a thin protective coating capable of mechanical or chemical removal. Unless otherwise specified, the fiber termination shall be unconnectorized with the end face suitably finished.

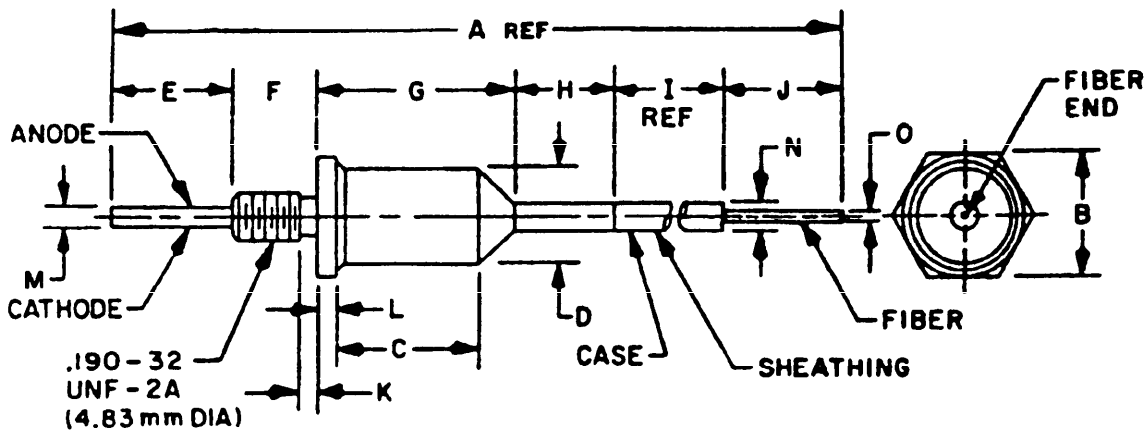
3.4 Performance characteristics. Specific performance levels required of each source type to satisfy the burn-in and qualification inspection requirements are as specified in tables III, IV, V, and VI.

3.5 Mass. The source mass (pigtail included) shall be a minimum consistent with the packaging design employed, and shall not exceed 5 grams.

3.6 Marking.

3.6.1 Identification. Marking size for source type-designation shall be 3 mm in height, and the manufacturer's designating symbol shall be identifiable without need for visual magnification.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T23:24Z
Check the source to verify that this is the current version before use.



Type No.	Letter Symbol	Dimensions				Note 1 applies to all
		MILLIMETERS		INCHES		
		Min	Max	Min	Max	
A08B B08B	A	349.3	362.0	13.75	14.25	
	B	11.0	11.25	.433	.443	
	C	13.0	13.5	.51	.53	
	D	8.4	8.9	.33	.35	
	E	10.0	10.2	.39	.40	
	F	7.4	7.9	.29	.31	
	G	20.0	20.3	.78	.80	
	H	7.6	7.9	.30	.31	
	I	288.93	295.28	11.375	11.625	
	J	10.06	10.11	.396	.398	5
	K	1.14	1.19	.045	.047	
	L	1.96	2.01	.077	.079	
	M	1.96	2.01	.077	.079	3
	N	2.3	2.5	.09	.10	4
	O	0.124	0.127	.0049	.0050	5
A08C B08C	A N	Same as above				
	O	0.150	0.155	.0059	.0061	

NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are given for general information only.
3. Anode and cathode conductors shall be insulated from each other. Dimension shown is the conductor's center-to-center spacing.
4. Case dimension only shown. Specified strain relief member affixed at this location (N) shall increase N to no more than 5.0 mm.
5. Optic fiber end face shall be suitably terminated (finished) within the J dimension.

FIGURE 1. LED sources package assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:24Z
Check the source to verify that this is the current version before use.

TABLE II. Primary LED source performance characteristics.

Sealing level	Common characteristics							Source dash number	Specific characteristics						
	Electrical					Optical			Pigtail/Optical				E/O System		
	I_F (mA)	V_F (V)	P_O 8/ 9/ (VT)	V_R 1/ (V)	R_θ 2/ (C/W)	λ_p 3/ (N.m)	$\Delta\lambda$ 4/ (N.m)		Fiber size (μM)	Core Index Type	NA	P_O (μW)	t_r/t_f 5/ (ns)	Band-width 6/ (MHz)	Line-arity 7/ (-dB)
Fully hermetic	100	1.5	0.15	3 V at 10 μA	40	800, 850, 900	40	A08B	50/125	GRADED	0.15	25	10	20	30
								A08C	100/140	STEP	0.3	100	10	20	30
Non-hermetic	100	1.5	0.15	3 V at 10 μA	40	800, 850, 900	40	B 08 B	50/125	GRADED	0.15	25	10	20	30
								A 08 C	100/140	STEP	0.3	100	10	20	30

1/ V_R is taken at a reverse current (I_R) equal to 10 μA.
2/ Thermal resistance (R_θ) is the junction to ambient value.
3/ λ_p is the wavelength at peak output amplitude.
4/ $\Delta\lambda$ is the spectral bandwidth between half amplitude points.
5/ t_r/t_f values are taken at the 10 and 90% amplitude points.
6/ Bandwidth value corresponds to the -3 dB power output level.
7/ Linearity level is value relative to fundamental level.
8/ Source detail specification P_O shall not decrease more than 50% at 105°C at 100 mA drive.
9/ The reliability of the device shall insure operation within all performance specifications, except P_O, for 10^7 hours at 25°C. At the end of 10^7 hours the power output (P_O) is allowed to be no less than 50% that specified in this table where the drive current has remained at 100 mA.

Source: https://assist.dla.mil -- Downloaded: 20260-19T23:24Z
Check the source to verify that this is the current version before use.

3.6.2 Safety. Sources covered by this associated detail specification are not expected to present a potential personnel hazard and no special safety marking is required.

3.7 Workmanship. Each source submitted for qualification shall be visually inspected. Sources shall show no evidence of excessive stress in the pigtail or strain-relief region such as, poor attachment, crimp or pinch conditions.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. Quality assurance provisions shall be in accordance with DOD-S-24622, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the inspections specified in table III (screening) and tables III, IV, V, and VI.

4.2.1 Screening inspection. Process conditioning and screening inspection shall be conducted on 100 percent of the lot prior to submission of the lot to the tests specified in tables IV, V, and VI herein. Table III lists the parameters with their interim and end point limits (for steady state power burn-in) which must be met for source acceptance. Tests shall be accomplished in the sequence shown in table III. End point parameters shown in table III are either of the maximum limit type or of the variational (delta, Δ) type. In particular, the parameters of spectral bandwidth ($\Delta\lambda$), data rate (PPS), frequency response (f) and signal linearity (L) must not be greater or less than (as specified) the limiting value noted. The other end-point parameters permit some variation between the pretest and post-test end-point readings and hence a delta level is specified for these. The symbol P_O, G, S represents the range of values of output powers acceptable from pigtails of the graded (G) or of the step (S) refractive index core type fiber, respectively.

4.3 Quality conformance inspection. Quality conformance inspection shall be in accordance with DOD-S-24622, and as specified herein.

4.3.1 Group A inspection. Group A inspection tests, LTPD's and performance limits are specified in table IV.

4.3.2 Group B inspection. Group B inspection examinations, tests, LTPD's and performance limits are specified in table V. Required end-point measurements are as specified following subgroups 2 and 3.

4.3.3 Group C inspection. Group C inspection, inspection tests, LTPD's and performance limits are specified in table VI. Required end-point measurements are as specified following subgroups 2, 3 and 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19491.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Lead length, material, finish if other than that specified, or when a choice is required by the detector application.
b. Pigtail fiber assembly length if other than that specified, or when a choice is required by the detector application.
c. Mounting arrangement to be used when unique to the intended detector application.

6.2 Engineering information (see table II). Typical primary electrical, optical and electro-optical performance characteristics (at T_A = 25°C) are shown on figure 2 for the LED source types covered by this associated detail specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:24Z
Check the source to verify that this is the current version before use.

TABLE III. Process conditioning and screening inspection.

Inspection	MIL-STD-750 Method	MIL-STD-750 Details	Symbol	Limits Min	Limits Max	Unit
High temperature life (nonoperating)	1031	Time = 24 hours T_A = +125°C (QLA) = +85°C (QLB)	---	---	---	---
Thermal shock (temperature cycling)	1051 (107 of MIL-STD-202) Test condition A-1	$T_{A(min)}$ = -40°C $T_{A(max)}$ = +125°C (QLA) = +85°C (QLB) No special mounting condition	---	---	---	---
Constant acceleration	2006	---	---	---	---	---
High temperature reverse bias	1038 Test condition A	Time = 48 hours T_A = +100°C (QLA) = +70°C (QLB) V_R = 2.4 V dc	I_R	---	10	µA
Steady state power burn-in	1038		λp	-3	+3	Nm
Pre burn-in (interim) end points:	Test condition B	I_F = 100 µA T_A = +25°C V_R = -3 V.	$\Delta\lambda$ P_O G S NA	38 20 90 -5	42 30 110 +5	Nm µW µW %
Forward voltage Reverse current Thermal resistance	4011 4016 4081		V_F I_R R_θ PPS t_r & t_f f L	--- --- 35 40 --- 20 30	--- 10 45 --- 10 --- ---	V µA °C/W MBPS ns MHz dB
Burn-in (forward bias)		Time = 96 hours I_F = 150 mA T_A = +100°C (QLA) = +70°C (QLB)	--- --- --- ---	--- --- --- ---	--- --- --- ---	--- --- --- ---
Post burn-in (final) end points		I_F = 100 mA T_A = +25°C V_R = -3 V	λp $\Delta\lambda$ ΔP_O G S ΔNA	λp-2 --- --- --- ---	λp+2 45 -20 5	Nm Nm % %
Forward voltage Reverse current Thermal resistance	4011 4016 4081		ΔV_F ΔI_R ΔR_θ PPS $\Delta(t_r$ & $t_f)$ f L	--- --- --- 40 --- 20 30	±5 +100 ±20 --- 10 --- ---	% % % MBPS % MHz dB

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:24Z
Check the source to verify that this is the current version before use.

TABLE III. Process conditioning and screening inspection - Continued.

Inspection	MIL-STD-750 Method	MIL-STD-750 Details	Symbol	Limits Min	Limits Max	Unit
Hermetic seal	1071	QLA sources only	---	---	---	---
Fine leakage rate	Test condition H					
Gross leakage rate	Test condition C	QLA sources tested to entire condition C. QLB sources tested to gross leak step 1 with fluid temperature = 85°C	---	---	---	---
External visual exam	2071	Pigtail attachment	---	---	---	---

TABLE IV. Group A inspection.

Inspection	MIL-STD-750 Method	MIL-STD-750 Details	LTPD	Symbol	Limits Min	Limits Max	Unit
Subgroup 1							
Visual and mechanical examination	2071		5	---	---	---	---
Subgroup 2							
Optical power ouput		T_A = +25°C		P_O G	20	30	µW
				P_O S	90	110	µW
Radiation pattern		θ = 0°	5	NA	-5	+5	%
Forward voltage	4011	I_F = 100 µA		V_F	-5	+5	%
Subgroup 3							
Breakdown voltage	4021	T_{MIN} = -20°C T_{MAX} = +100°C (QLA) = +85°C (QLB) I_R = 10 µA dc	5	$V_{(BR)}$	-3	---	V dc
Subgroup 4							
Data modulation rate		T_A = +25°C		PPS	40	---	MBPS
Pulse rise time and fall time				t_r & t_f	---	10	ns
Frequency bandwidth				f	50	---	MHz
Signal linearity				L	30	---	dB

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:24Z
Check the source to verify that this is the current version before use.

TABLE V. Group B inspection.

Inspection	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1							
Solderability	2026		15	---	---	---	---
Resistance to solvents	1022			---	---	---	---
Subgroup 2							
Thermal shock (temperature cycling)	1051	$T_{A(MIN)}$ = -40°C $T_{A(MAX)}$ = +125°C (QLA) = +85°C(QLB) No special mounting					
Hermetic seal							
Fine leakage rate	1071, Test condition H	QLA sources only	10	---	---	---	---
Gross leakage rate	1071, Test condition C	QLA sources tested to entire condition C. QLB sources tested only to gross leak step 1 with fluid temperature = 85°C.		---	---	---	---
End points:		T_A = +25°C					
Optical power ouput				P_O G	20	30	µW
				S	90	110	µW
Forward voltage	4011	I_F = 100 mA dc	10	V_F	---	2.0	V
Reverse current	4016	V_R = -3 V dc		I_R	---	10	µA
Subgroup 3							
Intermittent operation life (LTPD)	1037	T_A = +25°C I_{FM} = 150 mA PPS = 1 MBPS Duration = 24 hrs. Normal stud mount.	5	---	---	---	---
End points: Same as subgroup 2 and subgroup 4 of group A inspection							

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:24Z
Check the source to verify that this is the current version before use.

2 HOLES .156 DIA THRU (2 PLCS) / LETTER AT LIMIT

TABLE V. Group B inspection - Continued.

Inspection	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 4							
Decap internal visual design verification	2075		15	---	---	---	---
Bond strength	2037			---	---	---	---
Subgroup 5							
Thermal resistance	4081	T_2 = +100°C (QLA) = +70°C (QLB); T_1 = +25°C; I_F = 100 mA dc	10	R_θ	35	45	°C/W
Subgroup 6							
High temperature life (nonoperating) LTPD	1032	Time = 24 hours; T_A = +125°C (QLA) = +85°C (QLB)	10	---	---	---	---

TABLE VI. Group C inspection.

Inspection	MIL-STD-750		LTPD
	Method	Details	
Subgroup 1			
Physical dimensions	2066		15
Subgroup 2			
Thermal shock (glass strain)	1056 Test condition A	T_{FLUID} = +100°C (QLA), = +85°C (QLB)	
Terminal strength Tensions	2036 Test condition A	Electrical leads Tensile load = 5 newtons for 5 minutes, Fiber pigtail Tensile load = 20 newtons for 5 minutes	
Torque	2036 Test condition D1	Electrical leads Torque load = 0.1 newton-meter (1 lb-inch) for 10 seconds Fiber pigtail Torque load = 0.2 newton-meter (2 lb-inch) for 10 seconds	10
Hermetic seal Fine leakage rate	1071, Test condition H	QLA sources only	

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:24Z
Check the source to verify that this is the current version before use.

TABLE VI. Group C inspection - Continued.

Inspection	MIL-STD-750 Method	MIL-STD-750 Details	
Subgroup 2 - Continued			
Gross leakage rate	Test condition C	QLA sources tested to entire condition C, QLB sources tested only to gross leak step 1 with fluid temperature = 85°C	
Moisture resistance	1021		
External visual inspection	2071		
End points: Same as subgroup 2 of group B inspection			
Subgroup 3			
Shock	2016		
Vibration (variable frequency)	2056		10
Constant acceleration	2006		
End points: Same as subgroup 2			
Subgroup 4			
Salt atmosphere (corrosion)	1041		15
Subgroup 5			
Intermittent operation life (λ)	1036	T_A = +25°C, I_F = 100 mA, PPS = 1 MPPS, Duration = 96 hours Normal stud mounting	λ=10
End points: Same as subgroup 2 of group B inspection and subgroup 4 of group C inspection.			

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:24Z
Check the source to verify that this is the current version before use.

Custodian:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Navy - AS, EC
Air Force - 11, 17, 19 99
DLA - ES

User activity:
Navy - CG, MC

Agent:
DLA - ES

Preparing activity:
Navy - SH

(Project 6030-0012-1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:24Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Naval Sea Systems Command
Washington, DC 20362

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commander
Naval Sea Systems Command
ATTN: SEA 55Z3
DOD Standardization Program and Documents Division
Department of the Navy
Washington, DC 20362

Source: https://assist.dla.mil -- Downloaded: 202640-19T23:24Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-S-24622/1	Sources, LED, Fiber Optic, 820-910 Nanometers Wavelength Range, Glass Pigtailed Type (Metric)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:24Z
Check the source to verify that this is the current version before use.